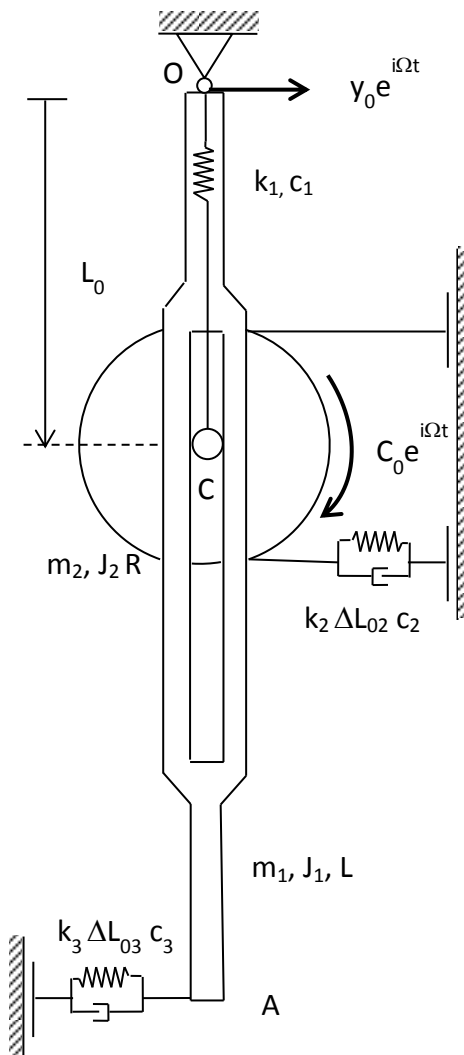


# Mechanical Systems Dynamic

December 12<sup>th</sup>, 2015 – Version A



|                           |                                   |                                  |
|---------------------------|-----------------------------------|----------------------------------|
| $m_1 = 10 \text{ kg}$     | $k_1 = 1000 \text{ N/m}$          | $y_{02} = 0.01 \text{ m}$        |
| $m_2 = 5 \text{ kg}$      | $k_2 = 2000 \text{ N/m}$          | $y_{03} = 0.02 \text{ m}$        |
| $J_1 = 0.8 \text{ kgm}^2$ | $k_3 = 3000 \text{ N/m}$          | $C_{01} = 2 \text{ Nm}$          |
| $J_2 = 0.2 \text{ kgm}^2$ | $\Delta L_{02} = 0.125 \text{ m}$ | $\varphi_1 = \pi/6 \text{ rad}$  |
| $L = 1.0 \text{ m}$       | $\Delta L_{03} = 0.1 \text{ m}$   | $\varphi_2 = -\pi/6 \text{ rad}$ |
| $L_0 = 0.6 \text{ m}$     | $c_1 = 2 \text{ Ns/m}$            | $\varphi_3 = \pi/3 \text{ rad}$  |
| $R = 0.3 \text{ m}$       | $c_2 = 4 \text{ Ns/m}$            | $\Omega_1 = 30 \text{ rad/s}$    |
|                           | $c_3 = 6 \text{ Ns/m}$            | $\Omega_2 = 20 \text{ rad/s}$    |
|                           |                                   | $\Omega_3 = 10 \text{ rad/s}$    |

The mechanical system in the picture lays in the vertical plane and is shown in its equilibrium position. It is constituted by a rigid bar (the center of gravity is in the middle bar length  $L$ , namely at  $L/2$  from point  $O$ ) and by a disk which is sliding and rotating with respect to the first body. The disk center is connected with point  $O$  with a spring and damper system ( $k_1, c_1$ ). The static preload of the springs  $k_2$  and  $k_3$  is given and is equal to  $\Delta L_{02}$  and  $\Delta L_{03}$  respectively. In the static configuration the distance between the center of the disk and point  $O$  is  $L_0$ .

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range  $0 - 10 \text{ Hz}$ , step =  $0.01 \text{ Hz}$ ) applying the torque  $C_0 e^{i\Omega t}$  at the disk. Output: the absolute rotation of the disk and the horizontal displacement of point  $C$ . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range  $0 - 10 \text{ Hz}$ , step =  $0.01 \text{ Hz}$ ) applying the horizontal displacement  $y_0 e^{i\Omega t}$  to the hinge  $O$ . Output: the vertical displacement of point  $C$  and the force transmitted by the spring and damper system ( $k_2, c_2$ ). Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the time history of the horizontal force applied to the hinge  $O$  given the following inputs: the horizontal periodic displacement  $y(t)$  defined by the equation below and a harmonic torque  $C_{01} e^{i(\Omega_1 t + \varphi_1)}$  applied to the disk. The time history shall be defined for one period of oscillation with a time step of  $0.001 \text{ s}$ . Save the results as SNxxx5.  

$$y(t) = y_{02} \cos(\Omega_2 t + \varphi_2) + y_{03} \cos(\Omega_3 t + \varphi_3)$$
6. Compute the static preload of the  $k_3$  spring applying a static preload of the  $k_2$  spring two times of the given  $\Delta L_{02}$  and keeping the same static configuration of the system. Write the numerical result on the sheets of the solution.

**Note:** Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select one of the following the folders *Test Dec. 12 2015 L07- A*, *Test Dec. 12 2015 L14- A*, *Test Dec. 12 2015 L15- A* according to the classroom and test version
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse...* (or *Sfoggia...*)
7. Fill the Title field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.